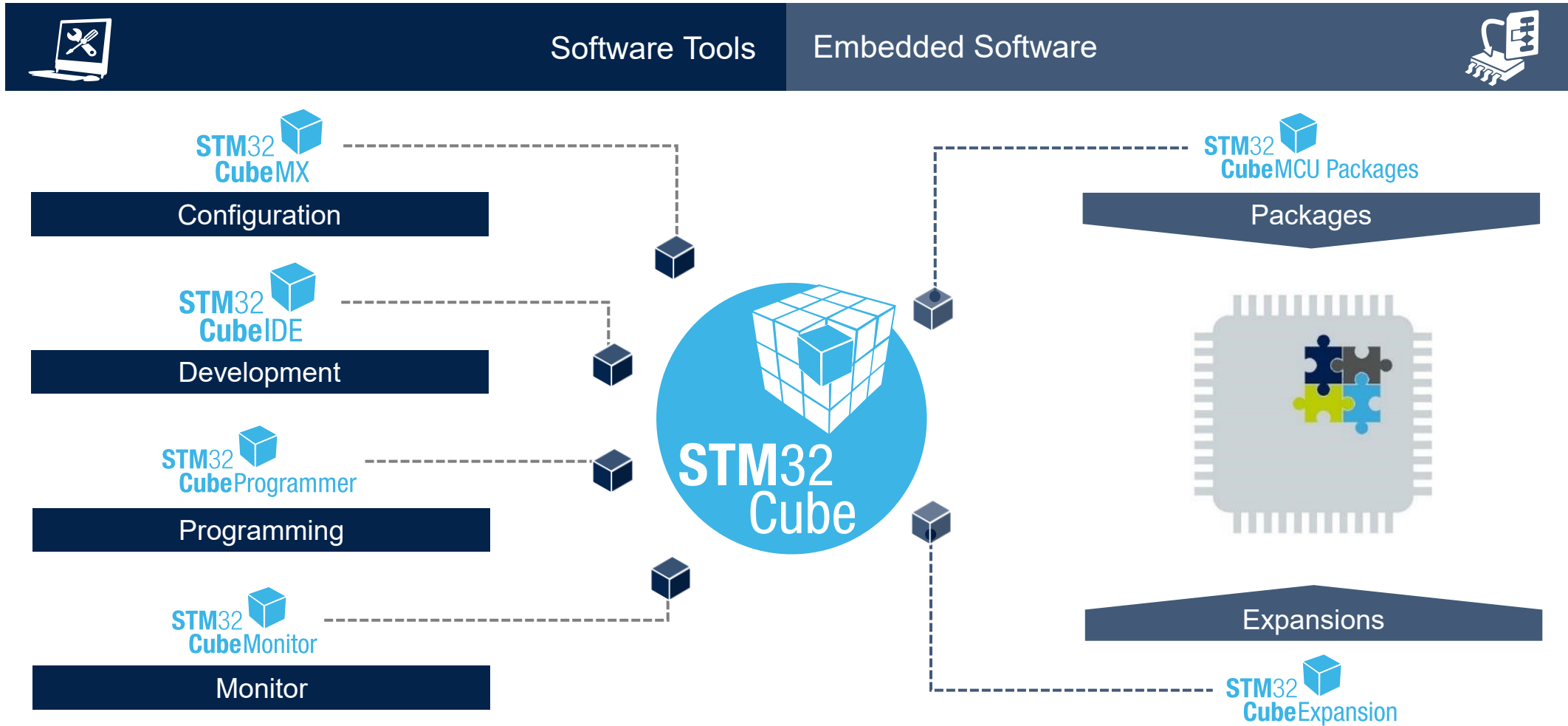




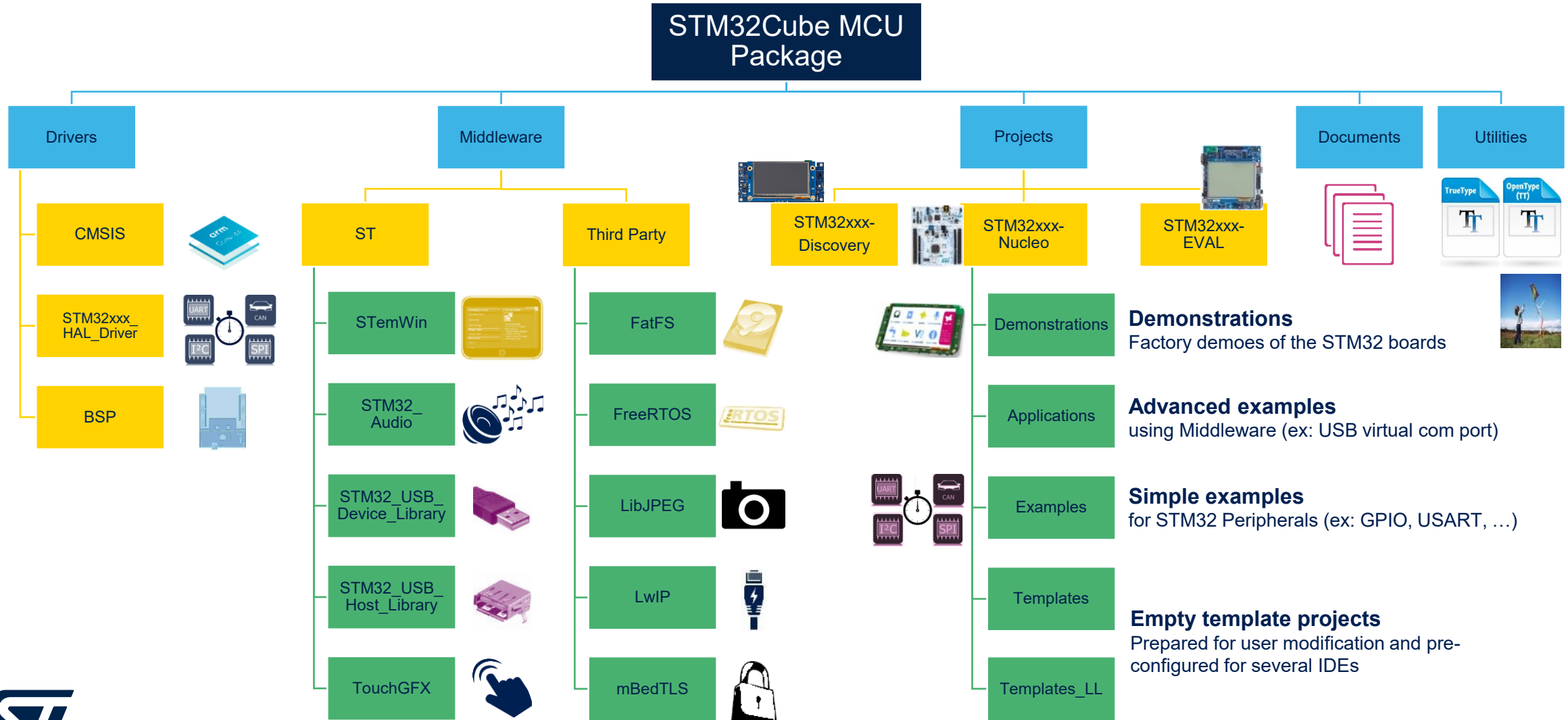
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# STM32Cube ecosystem Making STM32 development easier

# Inside the STM32Cube ecosystem

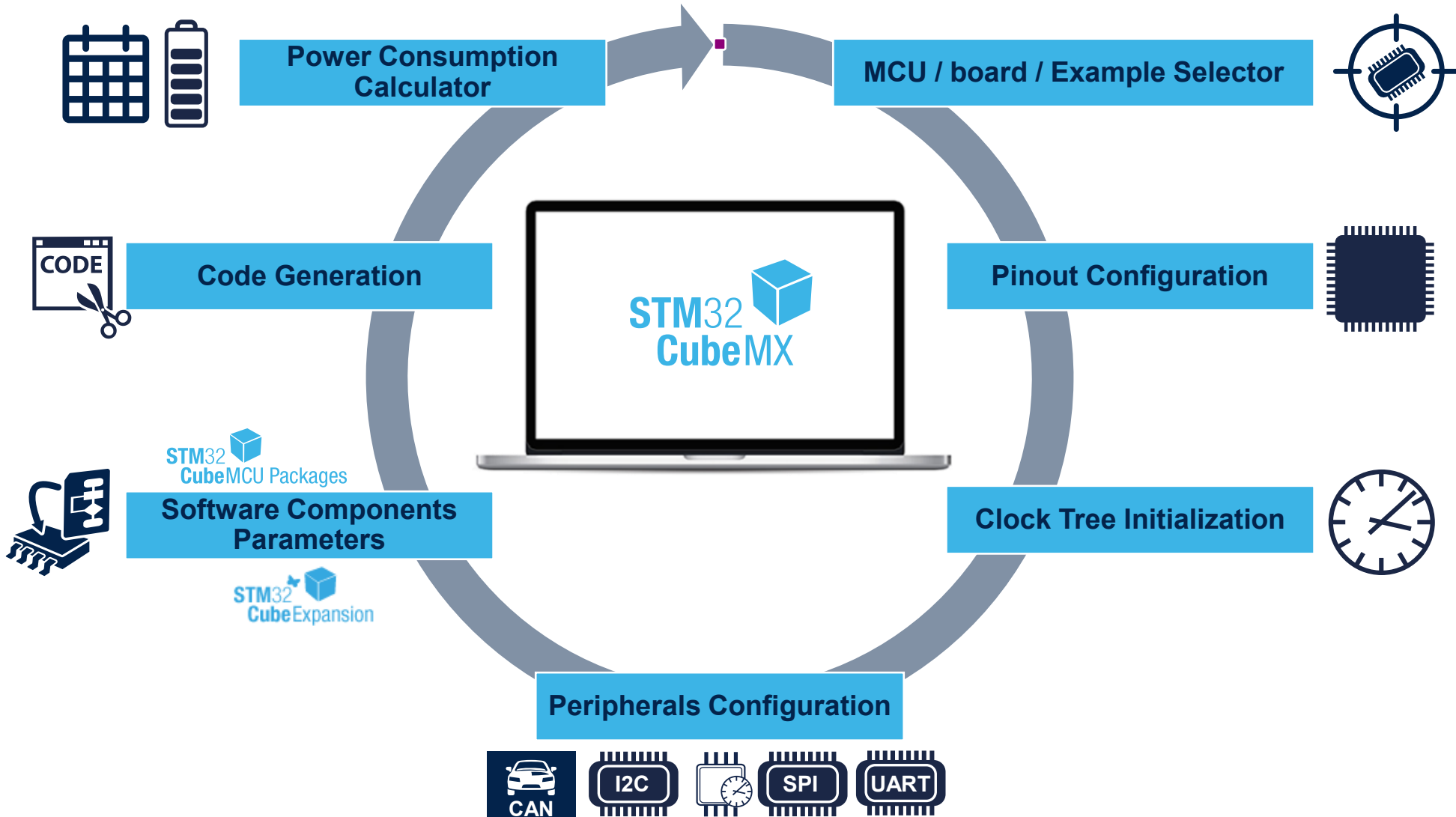


# Detailed content and organization



# STM32Cube Tools

# What is STM32CubeMX?

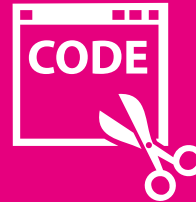


# What is STM32CubeIDE?

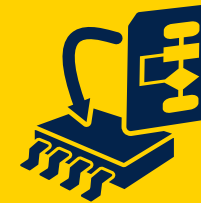
Chipset / Board  
Configuration



Code  
Development

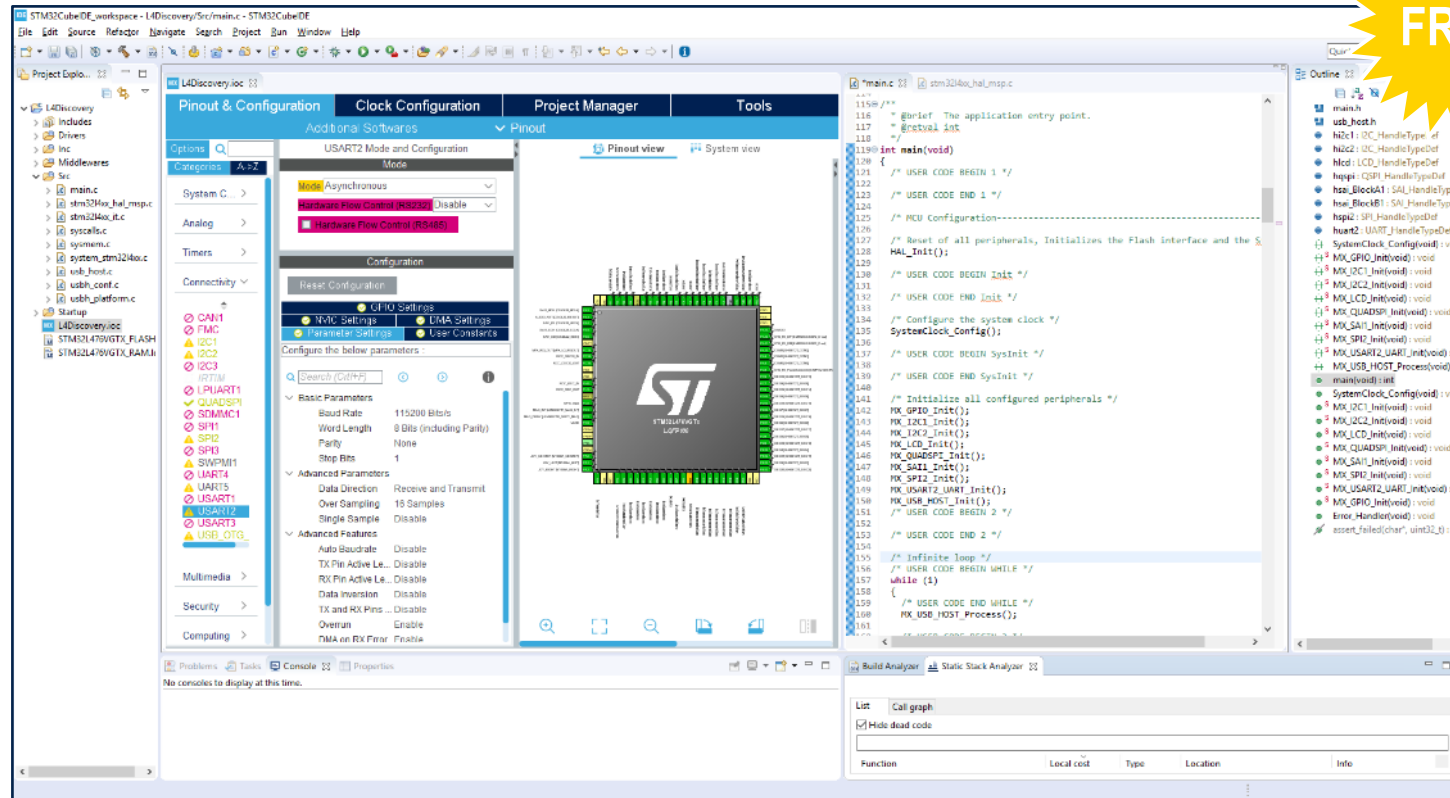


Validation  
Debug



# Free multi-platform development tool

FREE



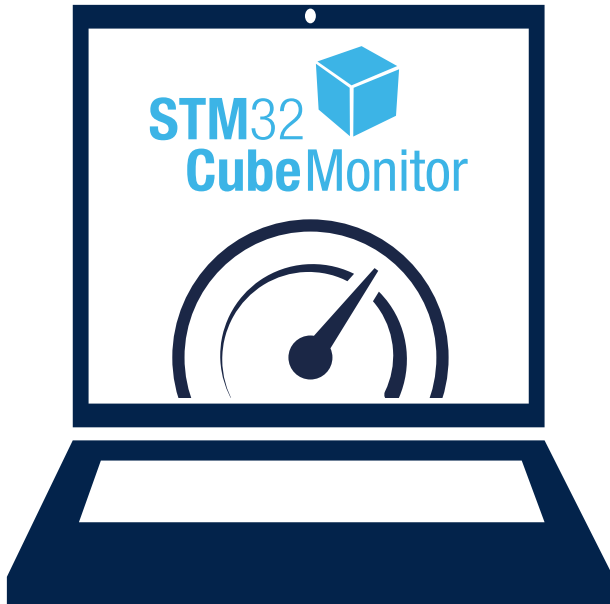
macOS

Eclipse/GCC Based

Free for Commercial Development

Multi-OS Support

# What is STM32CubeMonitor?



## Monitoring application variables during runtime

- Non-intrusive tool to follow application behavior without interruption.
- Real-time analysis to finetune application configuration.

## Drag & drop creation of dashboard UI

- Large choice of graphical components (gauges, bar graphs, plots...)
- Customize settings. No need for programming.
- Direct support of the Node-RED® open community.

## Graphical visualization on any display

- Multi-OS tool: direct support of PC, tablets and smartphones.
- Remote monitoring.

# Ecosystem 2021



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# STM32Cube Software Suite: Ongoing

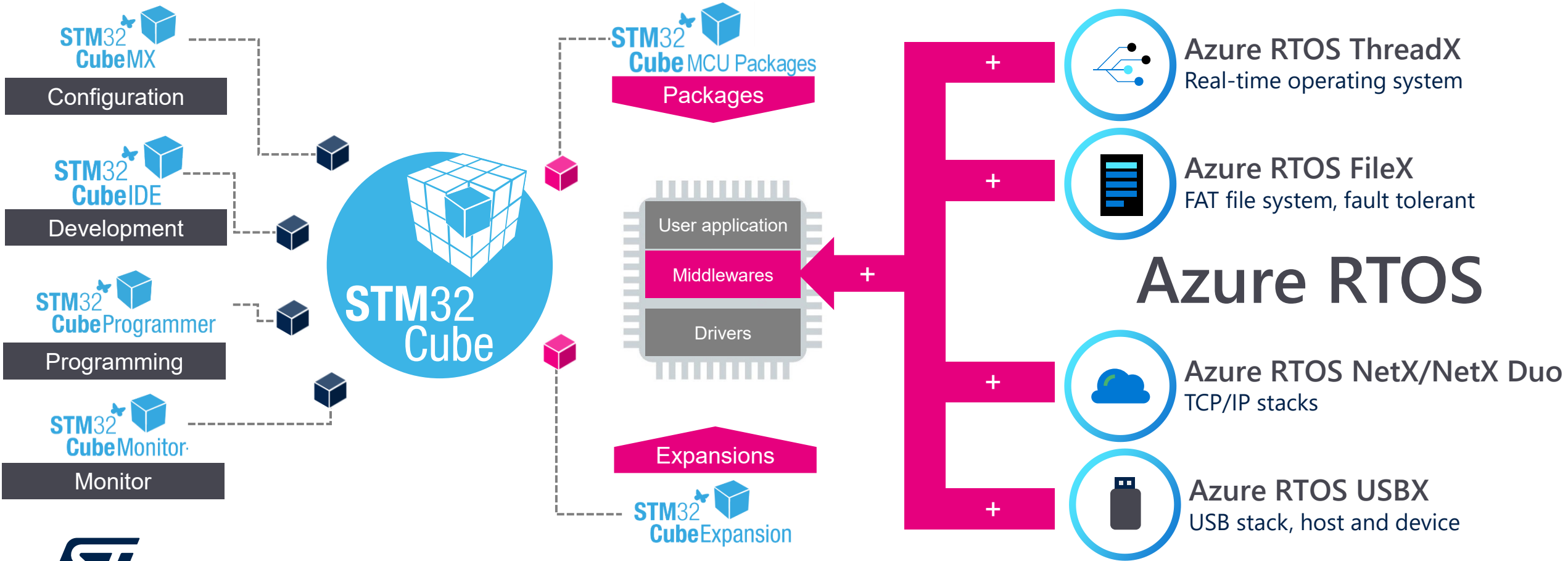
## 2021: Complemented with Microsoft AzureRTOS



Software Tools



Embedded Software





# STM32Cube Software Suite: Upcoming

Microsoft Azure RTOS bringing additional Key benefits to STM32Cube software Suite, from 2021



+ Azure RTOS

- Faster & Easier Development
- Business-friendly terms
- Better Quality

+

- Fast performance
- Complete consistent solution
- Industry certifications



# Azure RTOS Deployment within STM32 and STM32Cube portfolio



## Professional grade, highly reliable and market proven MW suite

- **Industrial grade** networking stack: optimized for performance coming with **many IoT protocols**
- Advanced FS/FTL: **fully featured** to support **NAND/NOR Flash** memories
- **USB Host** and **Device** stacks coming with **many classes**
- **Safety pre-certifications (from MSFT)**: IEC 61508 SIL4, IEC 62304 Class C and ISO 26262 ASIL D
- **Security pre-certifications (from MSFT)**: EAL4+ for TLS/DTLS, FIPS 140-2 for SW crypto lib

Azure RTOS  
Deployment

STM32H7

Q1

STM32F4  
STM32L4

Q2

STM32G4  
STM32F7  
STM32L5

Q3

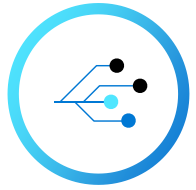
STM32WB  
STM32G0

Q4

STM32WL

Q4

2021



Azure RTOS ThreadX  
Real-time operating system

- **Small:** ~2KB Minimal Footprint
- **Fast:** Sub microsecond context switch, APIs
- **Safe:** SIL 4, ASIL D, Medical Class C
- **Security:** Extensive Pen Testing, Part of EAL4+, FIPS 140-2
- **Advanced:** Preemption-threshold, Event Chaining, Auto Scaling
- **Easy:** Consistent API, Extensive out-of-box examples, porting guideline from FreeRTOS available, CMSIS-OS layer

# Azure RTOS ThreadX

| Azure RTOS ThreadX API            |                    |
|-----------------------------------|--------------------|
| Thread Services                   | Messaging Queues   |
| Counting Semaphores               | Mutexes            |
| Event Flags                       | Block Memory Pools |
| Byte Memory Pools                 | Application Timers |
| Azure RTOS ThreadX Core Scheduler |                    |



Azure RTOS USBX  
USB stack, host and device

- **Small:** ~8.5KB Device, ~12KB Host
- **Fast:** Leverages DMA, Minimal Function Call Layering
- **Safe:** SIL 4, ASIL D, Medical Class C
- **Advanced:** Comprehensive class support
- **Easy:** Consistent API, Extensive out-of-box examples and device/host controller integration

# Azure RTOS USBX

| X USBX HOST API                          |                | X USBX DEVICE API              |
|------------------------------------------|----------------|--------------------------------|
| ASIX                                     | HUB            | CDC/ACM                        |
| AUDIO                                    | PIMA (PTP/MTP) | CDC/ECM                        |
| CDC/ACM                                  | PRINTER        | DFU                            |
| GSER                                     | PROLIFIC       | HID                            |
| HID                                      | STORAGE        | PIMA (PTP/MTP)                 |
| CDC/ECM                                  |                | STORAGE                        |
| USB HOST STACK                           |                | RNDIS                          |
|                                          |                | USBX DEVICE STACK              |
| OHCI, EHCI, PROPRIETARY HOST CONTROLLERS |                | PROPRIETARY DEVICE CONTROLLERS |



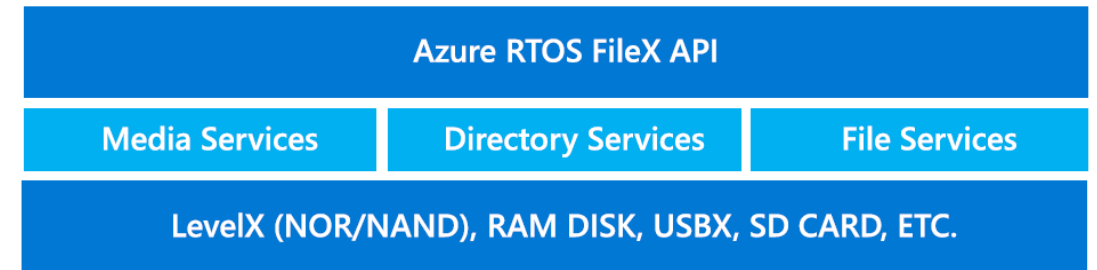
- **Small:** ~50KB Device-to-Cloud
- **Fast:** Near Wire Speed, Minimal CPU usage
- **Safe:** SIL 4, ASIL D, Medical Class C
- **Security:** Extensive Pen Testing, EAL4+, FIPS 140-2
- **Advanced:** Extensive Components, Zero Copy, Auto Scaling
- **Easy:** Consistent API, Extensive out-of-box examples

# Azure RTOS NetX Duo

|                                           |                 |                              |
|-------------------------------------------|-----------------|------------------------------|
| MQTT                                      | CoAP            | LwM2M                        |
| Auto IP                                   | HTTP, HTTPS     | SMTP                         |
| DHCP                                      | NAT             | SNMP                         |
| DNS, mDNS, DNS-SD                         | POP3            | Telnet                       |
| FTP, TFTP                                 | PPP, PPPoE      | PTP, SNTP                    |
| Azure RTOS NetX Duo                       |                 |                              |
| IGMP                                      | NetX Secure TLS | NetX Secure DTLS             |
| ICMP                                      | IPV4 & IPV6     | Azure RTOS NetX Secure IPsec |
| ARP/RARP                                  | UDP             |                              |
|                                           | 6LoWPAN         | TCP                          |
| Ethernet, Wi-Fi, Bluetooth LE, 15.4, etc. |                 |                              |
| Azure RTOS ThreadX                        |                 |                              |

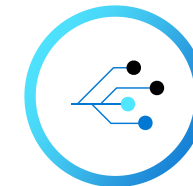


- **Small:** ~9KB Minimal Footprint
- **Fast:** Direct Data Write, Cache optimized for speed
- **Safe:** SIL 4, ASIL D, Medical Class C
- **Advanced:** Fault tolerant, FAT 12/16/32/exFAT, Extensive Cache Support, NAND/NOR Wear Leveling, Auto Scaling
- **Easy:** Consistent API, Extensive out-of-box examples



# X-CUBE-AZRTOS-H7

- AzureRTOS expansion package for STM32H7 series
- Fully hosted on ST Github account
- First release W51 2020
- Examples available for all middlewares
- Official release Q1 2021
- [Github.com/STMicroelectronics](https://github.com/STMicroelectronics)



**Azure RTOS ThreadX**  
Real-time operating system

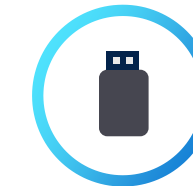


**Azure RTOS FileX**  
FAT file system, fault tolerant

## Azure RTOS



**Azure RTOS NetX/NetX Duo**  
TCP/IP stacks



**Azure RTOS USBX**  
USB stack, host and device

# Thank you

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